AETHER

Architecture & Decision Log

System design, the decisions taken and why, and the build plan to demo.

Project Autonomous Margin & Collateral Intelligence **Programme** GIFT IFIH Young Builders **Revision** B
Generated 2026-08-22

> **PROTOTYPE DOCUMENT**
> Describes a prototype under construction. All market data, counterparties, portfolios and legal documents referenced are **simulated** and clearly labelled as such in the repository. Nothing here asserts regulatory approval, certification under any industry methodology, or connectivity to any market infrastructure.

Working document. The narrative version, with diagrams and the full risk register, is the shared architecture artifact; this file is the one that gets edited during the build.

Scope: 48-hour solo prototype, demo 2026-08-16. One complete vertical slice, in depth, over breadth.

The slice

```
CSA document → Legal-Clause Agent → reviewed, frozen terms
             → Risk-Engine Agent → Collateral-Optimizer Agent → Decision → Audit
```

Three agents orchestrate; deterministic engines calculate. The boundary between them is a type, not a convention — see D-011.

Core principle

> **LLMs orchestrate. Deterministic engines calculate.**

Agents interpret, route, extract and orchestrate. Agents never perform authoritative financial calculations. A number crosses from the agent side to the engine side exactly once — through a typed schema, carrying source evidence, past a human confirmation when confidence is low.

Phase 0 results (complete)

CHECK	RESULT
Python	3.14.3 — **staying**, no downgrade needed
numpy / scipy	2.4.3 / 1.18.0, cp314 wheels confirmed. LP solver viable.
Outbound TLS	**Intercepted by Norton Web/Mail Shield.** Default context and certifi both fail.
TLS resolution	`truststore` + Windows trust store **works** — TLSv1.3 established
Demo mode	`live` available; `replay` **remains the default and the demo path**

CHECK	RESULT
Gate	15 passed, 3 warned, 0 failed

Decision D-001 — call `truststore.inject_into_ssl()` at API startup. Without it the Anthropic SDK cannot establish TLS on this machine. Guarded by `AETHER_USE_TRUSTSTORE`.

Decision D-002 — replay mode stays the demo path even though live now works. Live connectivity depends on a third-party AV product's certificate injection continuing to behave. That is not something to discover is broken during a judged presentation.

Decision log

#	DECISION	RATIONALE
D-001	`truststore.inject_into_ssl()` at startup	Only working TLS path on this machine
D-002	Replay is the demo path; live is optional	Removes network from the critical path
D-003	`Decimal` at every money boundary	Float drift in margin figures is indefensible and invisible
D-004	Single-curve discounting, no OIS/IBOR basis	Basis changes no conclusion in the demo; costs a day
D-005	SQLite, not Postgres	Docker is not installed. SQLite stays Postgres-shaped for later.
D-006	LP is continuous, not MILP	Solves exactly and instantly; integer lot sizes add no demo value
D-007	No vector DB / RAG	Document is bounded and small; full-document extraction is more accurate and removes a failure class
D-008	Markdown CSA fixture, not PDF	Text layer is what extraction consumes. PDF rendering is presentation, deferred until the UI exists.
D-009	Backend-first; frontend after the slice runs	An impressive parser with no engine behind it is worth nothing at the deadline
D-010	Cash carries the *highest* funding cost in the inventory	Reflects real opportunity cost, and is what makes the cash-first baseline genuinely expensive to beat
D-011	Agent tool-call arguments are `str | int | bool | None`	Makes "an LLM never produces an authoritative number" structural rather than procedural. A `Decimal` argument raises at construction, so a hallucinated figure cannot reach an engine even via a model-authored plan
D-012	Two IM figures, neither charged to the VM call	The schedule grid and a sensitivity aggregation disagree, and the disagreement is informative: a notional grid cannot see that a book is hedged. Charging either to a VM call would overstate the requirement
D-013	Sensitivities by full revaluation, not closed form	One repricing per bucket costs milliseconds and removes an entire class of analytic-derivative bugs
D-014	Settlement generates messages and never transmits	The Playbook's fourth feature is deliverable honestly as message *shape*. Connectivity is not, so it is refused rather than faked
D-015	Market data is ingested live and pinned before use	Satisfies "live market data" and "no calculation reads a feed" simultaneously. The network call happens in a script; the engine reads a pinned file

#	DECISION	RATIONALE
D-016	The optimizer only runs when the inventory owner is the Transferor	Optimising our own inventory against the counterparty's obligation is meaningless. When they owe, we issue a call and await their proposal
D-017	Two loaders, not one relaxed loader	`load_json` stays simulated-only so no real data can reach the engine by the demo path. Real published data has its own named door with four required conditions. Supporting ingestion never cost us the original guarantee
D-018	**No fixture value is changed after observing an output**	The rupee paper is not selected on the committed inventory, and that stands. A bill holding was briefly reduced to bring it back into the allocation; that was reverted, because tuning an input to produce a preferred result is the one thing a reproducibility claim cannot survive. The cross-border argument is carried by the ranking, which the optimizer computes for every eligible asset whether or not it is posted
D-019	The assistant is an agent whose tools cannot compute	"Ask AETHER" guides a user through the application, so it needs a model's flexibility. It does not need a model's arithmetic. Every tool selects named fields out of the deterministic run payload the panels render; none of them computes. The model chooses which to call and writes the prose between them
D-020	A live reply is verified against its tool results, and discarded whole if it fails	`guide.stray_figures` re-reads the generated prose and rejects any money, percentage or basis-point figure absent from the evidence behind it. The answer is not edited to remove the figure — a reply containing one invented number is not a reply to repair. The deterministic composition is served instead, and the response says so in `guard`
D-021	The assistant answers in replay with no model at all	A keyword router over the *same* tools composes from the same evidence, so the figures are identical either way and the demo needs no network. Live mode buys comprehension of unphrased questions and multi-tool answers, not different numbers
D-022	Retrieval moved out of the browser into the API	The previous assistant held eleven retrievers in TypeScript. Keeping them alongside the server-side tools would have created two sources for one figure. `frontend/src/lib/assistant.ts` was deleted rather than left as a fallback

Layering

```
backend/aether/core/      PURE. no I/O, no network, no clock, no ORM, no LLM.
backend/aether/agents/    LLM boundary: schemas, prompts, cache, replay.
backend/aether/api/       Thin routers. No business logic.
backend/aether/db/        Persistence.
```

Routers parse and validate, then call a service. Services orchestrate persistence and call `core/`. `core/` computes. Business logic never appears in a router; persistence never appears in `core/`.

Deterministic functions in the slice

Cut down from the full design to what the slice needs. All rows marked yes are built and tested.

#	FUNCTION	IN SLICE
F1	Discount curve, log-linear on DFs	yes
F2	Vanilla IRS PV, single curve	yes
F3	FX forward PV	yes

#	FUNCTION	IN SLICE
F5	Netting-set exposure, netting-set enforced	yes
F6	Credit Support Amount, MTA gate, rounding	yes — **the core**
F8	Asset haircut × FX haircut	yes
F9	Collateral optimization LP + baseline + duals	yes
F12	Audit hash chain	yes
F7	Initial margin, BCBS-IOSCO standardised schedule	yes — reported, **not** charged to the VM call
F13	Risk-weighted sensitivity margin	yes — bucketed delta by revaluation, own weights, **not SIMM**
F10	Settlement instruction, ISO 20022 / MT527 message shape	yes — generated, **never transmitted**
F4	FX option (Garman–Kohlhagen)	**cut** — and vega is therefore zero by construction
F11	Dispute attribution	**cut**

F6 — the calculation the product exists to get right

```
PostedValue         = Σ MV_a · valuation_pct_a · (1 − fx_haircut_a) · fx(a → base)
CreditSupportAmount = max(0, IA + max(0, Exposure) − Threshold)
DeliveryAmount      = CreditSupportAmount − PostedValue        if > 0
ReturnAmount        = PostedValue − CreditSupportAmount        if > 0

gate    if |Amount| < MTA → 0          ← tested on the UNROUNDED amount, per CSA 3(b)
round   ROUND_UP for delivery, ROUND_DOWN for return, to USD 100,000
```

F9 — the commercial thesis

```
minimise    Σ cost_a · x[a,c]
subject to  Σ x[a,c] · valuation_pct_a · (1 − fx_haircut_a) ≥ required_c
            Σ x[a,c] ≤ available_a
            x[a,c] = 0  if a ∉ eligible(c)          ← includes wrong-way exclusion
            Σ_{a ∈ class} x[a,c] ≤ cap_class
```

Three outputs beyond the allocation: - **Cost per effective unit** = `cost_a / (valuation_pct_a · (1 − fx_haircut_a))` — the cheapest-to-deliver ranking, and the reason the cheapest headline asset is not always chosen. - **Baseline comparison** against the cash-first heuristic. Measured, never asserted. - **Shadow prices** from the LP duals — "the binding constraint is the 20% equity cap; relaxing it by USD 1m saves N bp."

Build status — all blocks delivered

BLOCK	WORK	EXIT GATE	STATUS
1	`core/money.py`, `curves.py`, `pricing/`, `exposure.py`	Golden tests match analytically-derived values	done
2	`core/csa.py` (F6), `haircut.py` (F8), `im.py` (F7)	8 golden CSA cases green incl. MTA-before-rounding	done
3	`core/optimize.py` (F9), baseline, duals	`optimizer ≤ baseline` invariant green; wrong-way excluded	done

BLOCK	WORK	EXIT GATE	STATUS
4	`core/audit.py` (F12), decision, pipeline runner	Two runs, identical hashes. Slice runs headless.	done
5	Legal-Clause Agent: schema, prompt, replay, review gate, eval	Eval meets targets offline; freeze refuses unreviewed input	done
6	FastAPI routers, SSE, OpenAPI	Slice runs over HTTP	done
7	Frontend: desk, review, optimizer, audit, methodology	Demo runs in a browser	done
8	Demo scenarios, talk track, README	Full run offline	done
9	Playbook alignment: agent swarm, sensitivity margin, settlement, gold, cross-border surfacing, PDF and market-data ingestion	Agent boundary enforced by type and asserted by test; frozen terms govern the calculation	done

Final state: 406 tests passing, `ruff` and `mypy --strict` clean across 34 source files, frontend `tsc` and `eslint` clean, production build green across 7 routes.

Verified on the committed fixtures: net exposure (USD 19,469,510.12) → Credit Support Amount USD 9,469,510.12 → Delivery Amount USD 9,500,000.00; optimiser USD 5,496.52 against a cash-first baseline of USD 23,002.41, a 76.10% saving across five assets with four binding constraints; audit chain of 15 events — alternating agent decision and engine computation — verifying and reproducing byte-identically across runs.

Cut, as planned

F4 FX option and F11 dispute attribution. F4's absence is why vega is zero by construction rather than by omission: there is no instrument in the book whose value depends on a volatility, and the aggregation carries the term and reports zero with that reason attached.

Claims discipline

DO NOT WRITE	WRITE INSTEAD
"implements ISDA SIMM"	"BCBS-IOSCO schedule, plus a sensitivity aggregation using our own uncertified risk weights. Neither is SIMM"
"regulatory-compliant"	"designed against IFSCA / BCBS-IOSCO margin concepts"
"connects to SWIFT"	"generates ISO 20022 and MT527-shaped instructions; nothing is transmitted, no network exists"
"AI-calculated margin"	"AI-extracted terms, deterministically calculated margin"
"real-time market data"	"published data ingested once and pinned; no calculation reads a feed"
"zero errors"	"N golden cases and M invariants, all green; here is the eval score"

All demo data is simulated and labelled as such in every fixture's `_provenance` block.